

QRiousRAG: 모호한 질의응답에서 LLMs의 질의 정제를 통한 검색 증강 생성

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QRiousRAG: Retrieval Augmented Generation with Question-refinement using LLMs in ambiguous question answering

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Abstract

Although the question-answering performance of Retrieval-Augmented Generation (RAG) systems has improved dramatically with recent advances in language models, existing RAG systems still face challenges when providing accurate answers to ambiguous questions in open-domain question-answering tasks. In this paper, we propose the framework QRiousRAG suitable for ambiguous question-answering by applying retrieved document evaluation and informed, relevant, and curious question-refining using Large Language Models (LLMs). Our experimental results on the ASQA dataset demonstrate that our approach outperforms SOTA models, with 36.5% in ROUGE-L, 46.9% in Disambig-F1, and 41.3% in DR-Score.

1. Introduction

Large language models (LLMs) have shown powerful performance in various applications in natural language processing (NLP) and have dramatically improved the performance of question-answering (QA) systems [1]. In particular, retrieval augmented generation (RAG) systems, which combine information retrieval and generation tasks, have performed well [2]. However, since the answers in RAG systems essentially depend on the input questions, they still have difficulty in deriving consistent or reliable answers to ambiguous questions [3, 4]. This problem is especially prominent when information is incomplete or allows for multiple interpretations. To answer such a question extensively, it is necessary to explore past the first best answer.

In this study, we propose the new framework QRiousRAG utilizing LLM question-refinement to address this problem. The proposed methodology supplements the commonly used RAG approach with several specialized modules, more precisely we evaluate the relevance of retrieved documents and generate refined questions based on the refined documents and the initial query. With this it is possible to explore the solution space of the ambiguous question extensively, resulting in a better answer. Our model outperforms SOTA models and improves the overall score by 2.6 points compared to the baseline, demonstrating the superiority of our approach in ambiguous question answering.

The contributions of this paper are as follows.

- We propose a novel RAG model called QRiousRAG, which is specialized for ambiguous question-answering tasks.
- QRiousRAG extends the limitations of the existing QR method that excludes interaction between documents and queries.
- Our methodology outperforms SOTA models with 36.5% in ROUGE-L, 46.9% in Disambig-F1, and 41.3% in DR-Score.

This work was supported by the National Research Foundation of Korea(NRF) grant funded by the Korea government(MSIT) (RS-2021-NR066188).

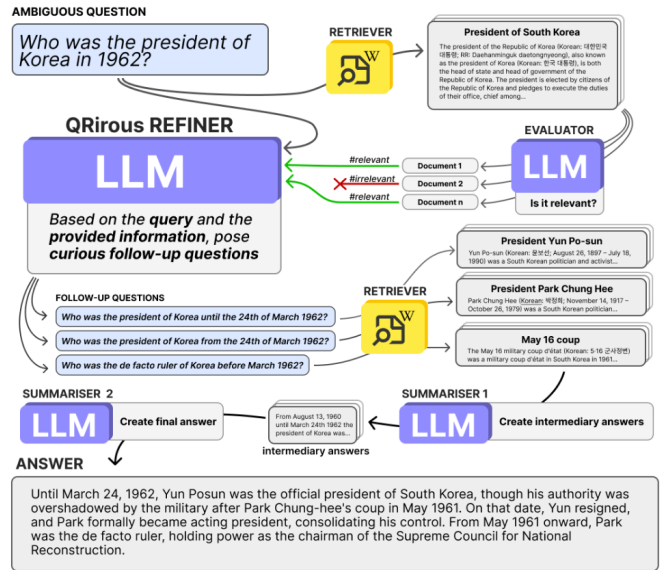


Figure 1. It shows the model structure and operating sequence of QRiousRAG framework. LLM's self-feedback is used to refine documents, refine questions through these, and create new questions. This is the process of outputting final answers based on new questions.

2. Related Work

In Open Domain Question Answering (ODQA) [3, 5], users often present ambiguous questions (AQ), which is a major obstacle for the Retrieval-Augmented Generation (RAG) system to provide accurate answers. In particular, it has been difficult to objectively measure the accuracy of answers to ambiguous questions. ASQA (Answer Summaries for Questions that are Ambiguous), proposed by Stelmakh et al. [4], provides a reliable evaluation criterion for AQ tasks and is designed to effectively evaluate answers based on

various interpretations of ambiguous questions. A recent study is the Tree of Clarifications (ToC) proposed by Kim et al. [6], which explores the AQ problem by generating various interpretations of ambiguous questions in a tree structure, with each branch handling different interpretations.

Query Rewriting (QR) is a widely used technique in information retrieval systems to improve retrieval performance by transforming users' queries. Ma et al. [7] improved performance in open-domain query answering by rewriting queries through the 'Rewrite-Retrieve-Read' framework, and Ye et al. [8] proposed a method to transform context-dependent queries into independent queries in conversational retrieval. However, existing QR approaches have limitations in actively reflecting the context of retrieved documents. To address these limitations, QRiousRAG utilizes the LLM self-feedback mechanism of Madaan et al. [9] to evaluate the relevance of retrieved documents and generate refined questions that reflect the initial query and the context of the documents based on the results. According to their study, LLM self-feedback can significantly improve the accuracy and reliability of answers, and QRiousRAG extends this to focus on reducing ambiguity within the RAG system. This goes beyond simple query modification and refines the solution space of ambiguous questions to provide a higher level of answer quality.

3. Methodology

This section gives a short task description followed by the structure of the proposed QRiousRAG framework.

Given is an ambiguous query q_{ambig} , which indirectly entails a set of unknown non-ambiguous queries $Q_{nambig} = \{q_1, \dots, q_i\}$, and a set of evidence documents $E = \{d_1, \dots, d_n\}$. The task is to use a subset of documents E_R that provides sufficient evidence to generate an answer $\hat{s}$ to the ambiguous query encompassing all solutions $S = \{s_1, \dots, s_i\}$ to Q_{nambig} .

The common RAG-approach is to retrieve E_R by selecting the top-k closest documents of the embedded q_{ambig} and each $d_i \in E$. Subsequently, the retrieved documents are provided to a generative model M that generates $\hat{s}$ based on in-context information.

QRiousRAG supplements this baseline approach with two additional modules: First, we append the evaluator M_{eval} to the retrieval stage to assess whether a retrieved document is relevant or irrelevant. Second, we propose a novel stage M_{Qrious} that generates a set of follow up questions $Q_{q-refine}$ trying to anticipate Q_{nambig} . The structure of QRiousRAG framework is shown in Figure 1.

3.1 Relevance Check

As shown in [9, 10, 11, 12], evaluating the retrieval results regarding their relevance for answering the query, increases model performance. We provided Q_{nambig} and $d \in E_R$ to the generative model M_{eval} and instructed it to output an evaluation token; *#relevant* for relevant documents and *#irrelevant* for irrelevant ones. We further include few-shot examples by adding several question/evidence-evaluation pairs as previous conversation to the prompt. Through this self-feedback of LLM, we redundant follow-up questions and the over-accumulation of evidence.

3.2 Question-refinement

The question refinement model M_{Qrious} is instructed to generate a set of new queries $Q_{refined}$ based on q_{ambig} and E_R . M_{Qrious} is

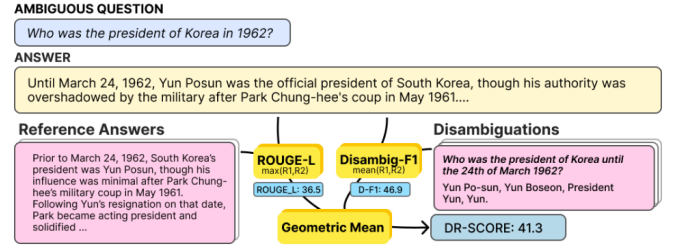


Figure 2. This is the evaluation process on ASQA dataset. For the generated answers, ROUGE-L and Disambig-F1 are calculated and the geometric mean of these is used to output the DR-Score.

explicitly prompted to come up with informed, relevant and curious follow-up questions that help answer q_{ambig} in a more holistic way. The positioning after the first retrieval stage is essential, as M_{Qrious} thus leverages the retrieved information to explore initially not covered aspects and explore retrieved information further. Furthermore, we provide few-shot examples to enforce a uniform output format for follow-up question extraction.

3.3 Structure Description

We briefly describe the whole QRiousRAG framework based on Figure 1. First, a question is given to retrieval and the Top-k documents are retrieved through Retrieval. Then, the retrieved documents are provided to the evaluation stage to assess document relevance. New questions are created from the refined documents and the original questions which are again provided for retrieval, where additional documents are retrieved for the new questions. New answers are created based on each follow-up question. Finally, the final answer is created considering all previous answers and evaluated.

4. Experiments

The experiments in this study focused on evaluating the self-feedback and question refinement process of LLM using ASQA benchmark.

4.1 Dataset

This study was evaluated on a dev-set subset of the ASQA benchmark. The ASQA benchmark is a subset of the AmbigNQ dataset [3], which includes Natural Questions [13], an open-domain QA. It consists of 6,316 samples with each one ambiguous questions, relevant evidence sources from Wikipedia, long-form answers, and Q&A pairs of implicit non-ambiguous queries. We scraped historical Wikipedia documents using the WikiMedia API, as some of the samples require time explicit information that changed since the dataset creation (e.g. who scored the most goal in international football?).

4.2 Experiment Setup

We use the NV-Embed-v2 [14] for the retrieval and embedding, and Llama-3.1 [15] as the generation model M . Both models are sourced from Huggingface Transformers [16] and quantized with BitsAndBytes [17]. The models were run on Nvidia A100 GPUs.

4.3 Evaluation

We measured the accuracy of the QRiousRAG framework with the DR-Score proposed by the ASQA benchmark. The DR-Score consists of ROUGE-L and Disambig-F1 metrics common problems of long-form QA assessment [18]. The structure of the evaluation metrics is shown in Figure 2.

Table 1. Evaluation metric for model performance on the ASQA

Model	ROUGE-L	Disambig-F1	DR-Score
ASQA (T5) [4]	39.2	26.4	32.1
ToC (GPT-3) [6]	39.7	33.7	36.6
Our models			
Baseline (Llama-3.1)	34.8	43.1	38.7
QRiousRAG (Llama-3.1)	36.5	46.9	41.3

ROUGE-L is calculated with the longest common sequence of a candidate, or generated answer, and the gold answer. As the ASQA benchmark provides two long-form answers per sample, a better score is used.

Disambig-F1 evaluates the fraction of disambiguated questions that can be answered from the predicted long answers to use Roberta (Liu et al. [19]) trained on SQuADv2. We apply the SQuADv2 model to the generated long answers to predict short answers to follow-up questions in the test dataset. We then compute the token-level F1 score between the predicted short answers and the ground truth short answers.

$$DR = \sqrt{\text{Disambig} - F1 * \text{ROUGE} - L}$$

Overall : The DR-Score is calculated as the geometric mean of Disambig-F1 and ROUGE-L.

4.4 Result Analysis

This section covers the accuracy measurement process of this methodology and the accuracy comparison with the baseline RAG system using ROUGE-L, Disambig-F1, and DR-Score.

Table 1 visually organizes the accuracy of models using ASQA dataset. The first model in the table is the best-performing model among the models presented in the ASQA paper. The second model in the table is the best-performing model among the models presented in the ToC paper. We measured the accuracy of the baseline RAG model and QRiousRAG framework. We found that QRiousRAG framework achieved the best performance of 46.9% in Disambig-F1 and 41.3% in DR-Score.

Effect of LLM's refinement: The new question $Q_{q-refine}$ generated through the question refinement process could generate more specific and clear questions than the existing ambiguous question q_{ambig} . In accuracy evaluation, the generated questions recorded a performance improvement of more than 1.7 in ROUGE-L, 3.8 in Disambig-F1, and 2.6 in DR-Score compared to the answers of the baseline RAG system. This means that the LLM's refinement improved the accuracy by generating more specific questions.

5. Conclusion

The methodology proposed in our paper has proven that the reliability of answers to ambiguous questions has been significantly improved through the self-feedback mechanism using LLMs in the RAG system. The experimental results on the ASQA dataset showed that the answer generation method, which incorporates document and question refinement of LLMs, achieved higher performance than the baseline RAG system. Additionally, it demonstrated excellent performance in deriving meaningful answers from ambiguous questions. Therefore, our study suggests the possibility of improving the accuracy of the RAG system through LLM's document and question refinement. We believe that it can provide a new solution to the ambiguous question-answering problem in the RAG system in the future.

We plan to evaluate our framework on a wider range of metrics and datasets. Through this, we will compare it with other models in more areas to find and supplement the shortcomings of QRiousRAG. We also plan to assess the fine-tuning of both the embedding/retrieval model and the question refinement model to increase its performance.

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